Aashray Jain

Software Developer | Full-Stack & AI/ML Engineer

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Summary

Software Engineer and Full-Stack & AI/ML Specialist with 4+ years of professional engineering experience across enterprise backend systems, scalable cloud architectures, and production LLM orchestration. Key contributor at Novafy rebuild of the NovaAgent v2 Python/FastAPI engine (190 commits) with MCP tool interfaces, prompt-caching (99% cached tokens), and ERP human-in-the-loop validation. Former TCS Systems Engineer with a proven track record delivering BFSI real-time risk platforms (500K+ users) and reducing cloud infrastructure costs by 70%. Inventor and author of ArihantQNet (novel tri-branch XAI diagnostic architecture) and officially registered copyright holder for ArihantQInvoice ERP. Currently pursuing M.Tech. in AI & Machine Learning at NIMS University Rajasthan.

Education

- **M.Tech. in Artificial Intelligence & Machine Learning** *2026 – 2028*
  NIMS Marik Institute of Computing, AI, Robotics & Cybernetics, NIMS University Rajasthan Pursuing
- **B.Tech. in Electronics & Communication Engineering** *2017 – 2021*
  Lakshmi Narain College of Technology, Bhopal, Madhya Pradesh (RGPV Bhopal) 8.74 CGPA
- **Senior Secondary** *2017*
  R.S. Memorial H.S. School, Katni, Madhya Pradesh (MP Board) 81.4%
- **Secondary** *2015*
  DAV ACC Public School, Katni, Madhya Pradesh (CBSE Board) 8.0 CGPA

Experience

- **Novafy OU** *Oct 2026 – Present*
  Software Development Intern Remote (Hamburg, Germany)
  - Core contributor to the ground-up rebuild of **NovaAgent v2 conversational engine** in Python, FastAPI, and PostgreSQL, authoring **190+ commits** through a reviewed PR engineering workflow.
  - Designed a modular plug-in LLM chat architecture with per-tenant tool configurations, capable of consuming and serving tools over the **Model Context Protocol (MCP)**, featuring dynamic multi-model fallback chains and database-driven model catalogs.
  - Engineered a **Human-in-the-Loop (HITL) safety layer** for Microsoft Dynamics 365 Business Central ERP integration; implemented pre-execution ERP metadata validation, one-click batch approval workflows, and value-substitution mechanisms preventing hallucinated or altered lookup codes.
  - Optimized latency and operational costs via provider-side **prompt-caching achieving ∼99% cached input tokens**; developed token-based conversation compaction and dynamic fit-to-window algorithms preventing context-overflow failures.
  - Built a comprehensive automated evaluation harness running 70+ conversational test scenarios to benchmark prompts and models; wrote unit and integration test suites utilizing isolated, throwaway PostgreSQL test containers.
  - Hardened system security with tenant and session isolation, replay protection, idempotent MS Teams redelivery handling, structured telemetry logging, and active outage metering/alerting.
- **ArihantQ** *Aug 2025 – Present*
  Founder (Part-Time and Experimental Venture) Remote (India)
  - Established a personal AI innovation lab focused on Generative AI, RAG, and LLM applications to build a portfolio of full-stack projects and strategically bridge a career transition.
  - Architected and developed key AI-driven prototypes, including an AI-Powered Portfolio WebApp and a RAG-Powered PDF Question Answering App, showcasing expertise in Python, HuggingFace, and LangChain.
  - Drove self-directed learning and validated technical concepts by actively participating in multiple hackathons (**Lyzr AI Architect Challenge, Elastic Forge the Future, MariaDB Python Hackathon**):
    - Achieved recognition in the Lyzr AI Architect Challenge, **winning a cash prize** for the innovative 'AI Cost Optimization Advisor' idea.
    - Attended the **Elastic Search AI company's virtual event in Bengaluru**, staying current with industry trends and networking.
    - An innovation idea in the MariaDB Hackathon received public recognition from the **MariaDB, MariaDB Foundation, CEO, and Community Advocacy**.
  - Conducted due diligence and secured business enablers (**Udyam Registration, Shop and Establishment Act License, and a Current Bank Account**) to provide a legal and professional foundation for an independent venture and continuous skill development.
  - Architected and developed ArihantQInvoice, a full-stack ERP platform for billing, inventory management, GST invoicing, stock tracking, purchase management, and customer/supplier operations using Node.js, Express.js, SQLite, and Sequelize.

- Registered software copyright for ArihantQInvoice Billing and Inventory Management with the Copyright Office, Government of India.
- **LinkedIn**: linkedin.com/company/arihantq

- **Tata Consultancy Services** *Jul 2021 – Jan 2025*
  Systems Engineer Mumbai & Kochi, India
  - Developed secure, real-time alert analysis systems and Line of Business (LOB) dashboards for a Canadian BFSI client using React, Node.js Lambda, and DynamoDB, improving risk monitoring efficiency by 90% and enhancing operational visibility.
  - Built and maintained scalable REST APIs and backend services using Node.js and Express.js to support secure financial workflows, modular integrations, and compliance-driven enterprise applications.
  - Led a multi-account AWS cloud migration for a large-scale enterprise platform, optimizing infrastructure architecture and reducing operational costs by 70% while improving scalability and resilience.
  - Strengthened enterprise security posture by integrating SAML-based authentication and AWS Cognito for secure user access across applications serving 500K+ users, reducing vulnerabilities and improving compliance by 40%.
  - Automated complex business reporting workflows using Angular, NX Monorepo, Python (Pandas, Flask), and Excel automation, reducing manual effort by 15+ hours per week and improving reporting accuracy.
  - Designed interactive analytics dashboards and visualization tools using Angular and D3.js to accelerate business intelligence and decision-making for enterprise stakeholders.
  - Collaborated with TCS Rapid Labs on AI/ML and automation PoCs, contributing to innovation initiatives in NLP, Computer Vision, and Generative AI for internal and enterprise use cases.
  - Built and evaluated machine learning pipelines for sentiment analysis and classification, strengthening capabilities in data-driven insights and intelligent automation.

- **Tata Consultancy Services** *Mar 2021 – Jun 2021*
  Project Intern Remote (India)
  - Developed the Asset Procurement & Inventory module for the EquipMax 3.0 project, utilizing Angular, Ionic, and Bootstrap as part of the TCS Accelerator brand project (EXOP-00937).

Projects

- **Class-Balanced Benchmarking & XAI Clinical DSS (ArihantQNet)** *Oct 2026*
  PyTorch, TorchVision, Python, Streamlit, Grad-CAM (XAI), Scikit-learn, OpenCV Live App | GitHub
  - Architected an end-to-end deep learning and Explainable AI (XAI) clinical intelligence framework for multi-stage Alzheimer's Disease stratification from 6,400 T1-weighted axial brain MRI scans (70/15/15 stratified partition).
  - Engineered a Normalized Square-Root Inverse-Frequency Weighted Cross-Entropy Loss with AdamW, compressing an extreme 49.78:1 class skew to 7.06:1 and achieving 100% sensitivity on the rare Moderate Demented class.
  - Benchmarked ResNet50 (88.44% accuracy, 0.8847 Weighted F1) and DenseNet121 (90.10% Balanced Accuracy, 100% recall on severe pathology with 3× smaller footprint: 28.4 MB vs 94.4 MB).
  - Formulated the novel tri-branch hybrid architecture **ArihantQNet**, fusing ResNet50 (macro-contextual key), DenseNet121 (fine-boundary query), and EfficientNet-B0 (channel-calibrated value) via 1×1 bottleneck projections, Quantum-Inspired Multi-Head Cross-Attention (Q-MCA), and Squeeze-and-Excitation (SE) recalibration.
  - Integrated Grad-CAM neuroanatomical heatmaps (hippocampus and ventricular atrophy) and deployed an interactive 6-module Streamlit clinical intelligence workbench for live inference and diagnostic reporting.

- **ArihantQInvoice Billing & Inventory Management System** *Apr 2026*
  Node.js, Express.js, SQLite, Sequelize, EJS, Swagger, DOCX Generation
  - Designed and developed a production-ready ERP and billing platform to digitize retail operations, replacing manual billing, stock registers, and purchase records with an integrated digital workflow for a retail textile business.
  - Implemented session-based authentication, GST and non-GST invoice workflows, automated DOCX invoice generation, dynamic pricing engine, and stock movement tracking across 50+ REST APIs and server-rendered modules.
  - Developed comprehensive OpenAPI/Swagger documentation for full API visibility and operational scaling.
  - Officially registered software copyright for ArihantQInvoice Billing and Inventory Management with the Copyright Office, Government of India.

- **AI-Powered Portfolio & Public Profile WebApp** *Jul 2025*
  HTML, CSS, JS, Bootstrap 5, GitHub Pages Project URL | GitHub
  - Developed a responsive, fully client-side portfolio to showcase full-stack, AI, and Quantum projects—no frameworks or backend required.
  - Integrated LinkedIn public badge sync, dark/light theme toggle, SEO OG metadata, and social profile links (GitHub, HuggingFace, Naukri, MyBharat, Credly).
  - Achieved zero-cost, CI/CD-ready deployment using GitHub Pages; optimized performance and mobile-first layout.

- **RAG-Powered PDF Question Answering App** *Jul 2025*
  Python, Streamlit, LangChain, HuggingFace, FAISS Live App | Hugging Face
  - Engineered a document Q&A application using Retrieval-Augmented Generation (RAG), enabling users to upload PDFs and retrieve answers in plain English.

  - Designed for model flexibility via HuggingFace (DeepSeek, Mixtral, Zephyr) and powered high-speed semantic chunk search using FAISS.
  - Created an intuitive dark-themed Streamlit UI with support for chat and text generation modes.

- **AI Cost Optimization Advisor (Lyzr AI Architect Challenge)** *Jun 2025*
  Lyzr Studio, AWS Bedrock, Claude, GPT, Gemini, Mistral Live Agent | GitHub
  - Developed a GenAI agent to solve the problem of LLM cost-effectiveness by analyzing AI use cases and recommending optimal models with ROI estimation and JSON insights.
  - Indexed 25+ knowledge sources with optimal chunking for RAG-based retrieval, including grounding facts to mitigate hallucination risks.
  - Simulated use cases (support triage, contract summaries), achieving a documented 666% ROI in the challenge.

Technical Skills

- **Programming Languages:** Python, JavaScript (ES6+), TypeScript, SQL
- **AI / ML & GenAI Frameworks:** FastAPI, Model Context Protocol (MCP), PyTorch, LangChain, Hugging Face, FAISS, Prompt-Caching, Streamlit, Grad-CAM (XAI)
- **Backend Architecture:** Node.js, Express.js, REST APIs, Microservices, PostgreSQL, SQLite, DynamoDB, Human-in-the-Loop (HITL) Workflows
- **Frontend Development:** ReactJS, Angular, HTML5, CSS3, Bootstrap 5, Material UI, D3.js
- **Cloud & Enterprise DevOps:** AWS (Lambda, API Gateway, S3, Cognito, DynamoDB, CDK), GCP (Vertex AI), Docker, Git, CI/CD, Swagger/OpenAPI
- **Enterprise Security:** SAML, AWS Cognito, CyberArk, RSA, Role-Based Access Control (RBAC), Replay Protection
- **Ecosystems & Integration:** Microsoft 365, MS Teams App, Microsoft Dynamics 365 Business Central, SharePoint, Outlook

Achievements

- **Top 100 Winner** — Microsoft AI Odyssey Challenge organized by Microsoft & TCS. *2024*
- **Azure AI Solution (AI-102T00)** — Completed 1920-minute advanced training program by Koenig Solutions. *2024*
- **Microsoft Learn Honors** — Earned 49 Badges, 10 Trophies, 48 Module completions, and 2 Applied Skills Certifications. *2024*
- **Hackathon Accolades** — Lyzr AI Architect Challenge Winner; public recognition from MariaDB CEO & Foundation. *2025*
- **National AI Recognition** — **FutureSkills Prime (AI Applications)** by NASSCOM & Govt. of India; MyBharat Honors. *2025*

Certifications

- **Cloud & AI Systems:** **Microsoft**: Applied Skills: GenAI with Azure OpenAI | **Microsoft**: Applied Skills: Document Processing with Azure AI | **AWS**: Cloud Practitioner Essentials (Second Edition)
- **Google Cloud Badges:** GenAI Apps with Gemini & Streamlit | Explore GenAI with Vertex AI Gemini API | Gemini Multimodality & Multimodal RAG | Text Prompt Engineering Techniques
- **Data Science & Foundations:** **IBM**: Applied Data Science with Python (Level 2) | **IBM**: Big Data Foundations (Level 2) | **IBM**: Artificial Intelligence Fundamentals | **IBM**: Data Fundamentals
- **Academic & Developer Credentials:** **IIT Kanpur (NPTEL)**: Programming in C | **IIT Kharagpur (NPTEL)**: Programming in Java | **HackerRank**: Java (Basic) Certificate & Gold Badge | **SEBI**: Investor Certification (ID: NISM20250000203277, Valid till Jul 2027)

Extracurricular & Volunteering

- Actively participate in GenAI hackathons, AI webinars, and developer communities; maintain personal discipline and endurance through competitive swimming and long-distance running.
- **Blood Donor** — Donated blood at TCS Blood Donation Camp, organized by Amrita Hospital, Kochi. *Feb 2023*
- **TEDxLNCTBhopalWomen** — Attended TEDxWomen event focused on empowerment and innovation at LNCT. *Nov 2017*
- **Planet Engineers & Convolution** — Participated in national-level technical fests and R&D workshops at LNCT and RGPV. *2018*